

Anh (Joe) Nguyen

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HIGHLIGHT

- I'm a PhD student working on Language-conditioned reinforcement learning (RL) agents: how to make AI agents understand human language and act accordingly, resulting in one publication at NeuRIPS [1] and some RL projects [5].
- I also have experience in 1) multimodal (visual+language) referring expressions: building the first underwater dataset and developing a method to clarify ambiguous expressions, and 2) NLP problems (text summarization, OCR, text classification).

SELECTED RESEARCH EXPERIENCE

Oregon State University — Research Asisstant

Jan 2022 - present

- LLM-based Computer use agent: Developing a GUI (computer) agent that can understand text and video tutorials to achieve a better success rate for daily-life computer tasks. Objective: better success rates in OSWorld.
- Language model-based RL agent: Developed an agent that can understand language descriptions of environment dynamics and act accordingly. The agent achieves state-of-the-art generalization over new language and unseen environment dynamics, increasing 6%-21% success rates over existing baselines (Dynamang, EMMA, LWM) [1]
- Visual language referring expression: Collected a new dataset in referring expressions for underwater objects. Compared and analyzed results from the state-of-the-art methods OFA and CLIP: both have low accuracy performance on underwater objects that are rare in territorial settings.
- *Advised by Prof. Stefan Lee*

Singapore Management University — Research Engineer

Nov 2019-Nov 2021

- Developed a method that can clarify ambiguous language queries in visual domains (referring expression): detect the most ambiguous portion of the query based on neural module network, then raise an relevant clarifying question to users.
- The method results in more efficient clarifying process: less than 10% of clarifying questions than existing baselines in CLEVR-Ref and Ref-Reasoning.
- *Advised by Prof Jiang Jing*

PUBLICATION

1. Anh Nguyen, Stefan Lee. [Language-conditioned world model improves policy generalization by reading environmental descriptions](#), accepted at Bridging Language, Agent, and World Models workshop at NeurIPS2025.
2. Anh Nguyen, Duy Tue Tran Van, Minh Quoc Nghiem. [Bachelor Thesis: Extractive summarization with bidirectional encoder representations from transformers](#): achieves state-of-the-art *long-text* summarization on CNN/Daily Mail (2019).
3. Duy Phung, Tu Minh, Anh Nguyen, Tien Dinh, “DTA Hunter System: A new statistic-based framework of predicting future demand for taxi drivers”, accepted for presentation @ SoICT 2017 (The Eighth International Symposium on Information and Communication Technology)

EDUCATION

- 2022 - 2027 (Expect) Msc ([thesis](#)) and PhD (Artificial Intelligence) at **Oregon State University**
GPA: 3.9/4.0
Advised by Professor [Stefan Lee](#)
- 2014 - 2019 Bachelor's Degree at **VNU-Ho Chi Minh University of Science**
(GPA: 3.8/4.0, top 5% of Department)
Advised by Dr. [Nghiem Quoc Minh](#)

ACADEMIC PROJECTS

- **Language-conditioned LLM-based world model**: Identified the research gap of current LLM-based model based RL systems: they failed to incorporate language into the world model, thus unable to change world modeling on the fly based on language.
- **Exploration in RL**: Compared different exploration strategies in model-based RL: count-based, curiosity-based and Monte-Carlo dropout in PointMaze
- **Search in Games (Sudoku and (M,N,K))**: Compared different tree search in Sodoku and (M,N,K) game: Monte-Carlo Tree Search, A* and Minimax with Alpha-Beta pruning.

AWARDS

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| 2025 | Scholarly Presentation Award from Oregon State University. |
| 2022 | Vietnam Education Foundation (VEF) 2.0 candidate for Ph.D programs in the USA. |
| 2019 | Fully funded exchange at APEC Digital Innovation and Entrepreneurship Forum (Taiwan) |
| 2019 | Fully funded exchange at GKS ASEAN SCIENCE at Kyungwoon University (Korea) |
| 2019 | Fully funded exchange at KAIST (South Korea) |
| 2019 | Fully funded exchange at Southeast Asia Machine Learning School |
| 2019 | Scholarship from Shinhan Bank for excellent students |
| 2017 | Scholarship from American Chamber of Commerce Vietnam for excellent students |
| 2017 | Scholarship from Global Cybersoft for excellent students |
| 2017 | Scholarship from Esilicon Vietnam for excellent students |
| 2016 | ACM-ICPC Vietnam National Competitive programming: Third Prize |
| 2016 | HCMc University of Science Olympiad in Informatics : Honorable mention |
| 2014 | Viethope Scholarship for excellent college students |
| 2013 | National Vietnam Olympiad in Informatics: Honorable Mention |
| 2013 | Vallet scholarship for excellent students |

SKILLS

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| Programming | C++ (expert), Python (expert), Java (familiar with), competitive programming |
| Research (expert) | Pytorch, Jax, Tensorflow, deep learning, reinforcement learning, LLM |